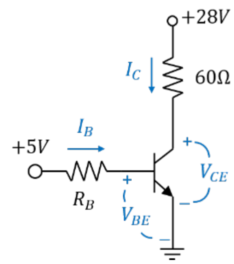
EE 4113 Midterm 2 Review Problems

From Lab Exercise 08 Prelab Question:

<u>BJT and the common emitter configuration</u>

1. The BJT in Figure 1 is in a forward-active region with $V_{BE} = 0.7V$, $I_B = 12mA$, and $V_{CE} = 2.5V$.



$I_C = \frac{28-2.5}{60} = 0.425 \rightarrow 425mA$

Figure 1: BJT in Common Emitter Configuration

a) What is the power dissipated in the BJT? $I_C = \frac{28-2.5}{60} = 0.425 = 425mA$

$$P_{BJT} = I_B V_{BC} + I_C V_{CE}$$
$$= (12\cdot10^{-3})(0.7) + (425\cdot10^{-3})(2.5) = \boxed{1.071W}$$

b) What is the power dissipated in the load?

$$P_{load} = I^2 \cdot R_L = (425\cdot10^{-3})^2\cdot(60) = \boxed{10.84W}$$

2. The BJT in Figure 1 is in a saturation region with $V_{BE} = 0.8V$, $I_B = 30mA$, and $V_{CE} = 0.3V$.

a) What is the power dissipated in the BJT?

$$P_{BJT} = I_B V_{BC} + I_C V_{CE} = \boxed{162.5mW}$$

b) What is the power dissipated in the load?

$$P_{load} = I_B V_{BC} + I_C V_{CE} = \boxed{12.79W}$$

From Lecture 7a:

$f = 100\,Hz$

Total Harmonic Distortion

3. An amplifier with an input signal $v_s(t) = 0.25\sin(2\pi 100t)\ Volts$ has the following output,

$$v_o(t) = 12.0 - \underbrace{7}_{A_1}\sin(2\pi 100t) + \underbrace{0.25}_{A_2}\sin(2\pi 200t - 30°) + \cdots$$
$$\underbrace{2.5}_{A_3}\sin(2\pi 300t - 70°) + 0.5\sin(2\pi 500t - 130°)\ Volts$$

$A_4 = 0$ $A_5 = 0$

a) What is the total harmonic distortion?

$$= 100\%\left[\frac{(0.25)^2 + (2.5)^2 + (0)^2 + (0.5)^2}{(7)^2}\right] = 100\%\left[\frac{0.625 + 6.25 + 0.25}{49}\right] = 100\%\left[\frac{6.5625}{49}\right] \approx \boxed{13.39\%}$$

4. An amplifier circuit has an input sine wave at 1 kHz, and the FFT of the output has the following (frequency, magnitude) pairs.

$(1\ kHz, +17\ dBV)$ a_1
$(2\ kHz, -4\ dBV)$ a_2
$(3\ kHz, +2\ dBV)$ a_3
$(4\ kHz, -8\ dBV)$ a_4
$(5\ kHz, 0dBV)$ a_5

$$A = 10^{\frac{a}{20}}$$

$$A_1 = 10^{\frac{17}{20}} = 7.079 = \boxed{7.08V}$$
$$A_2 = 10^{\frac{-4}{20}} = 0.63096 = \boxed{0.63V}$$
$$A_3 = 10^{\frac{2}{20}} = 1.25893 = \boxed{1.26V}$$
$$A_4 = 10^{\frac{-8}{20}} = 0.39811 = \boxed{0.40V}$$
$$A_5 = 10^{\frac{0}{20}} = \boxed{1}$$

a) What is the total harmonic distortion?

$$100\%\left[\frac{(0.63)^2 + (1.26)^2 + (0.40)^2 + 1^2}{(7.08)^2}\right] = 100\%\left[\frac{0.3969 + 1.5876 + 0.16 + 1}{50.1264}\right]$$

$$= 100\%\left[\frac{3.1445}{50.1264}\right] = 6.3\%$$

Lecture 7b instantaneous power and average power:

Power Calculations

5. While dB (decibel) is an expression for the gain (and is dimensionless/unitless), dBm (decibels per milliwatt) is a concrete measurement of the power level. The dBm conversion formulas are given as,

$$(y)\ dBm = 10\log_{10}\left(\frac{x\ mW}{1\ mW}\right)$$

And

$$(x)\ mW = (1\ mW)\left(10^{\frac{y}{10}}\right)$$

a) How many dBm is 150 microwatts?

$$150 \cdot 10^{-6}\ W = 150 \times 10^{-3} \underbrace{\times 10^{-3}\ W}_{mW}$$

$$0.15\ mW$$

$$dbm = 10\log_{10}(0.15) = -8.23908\ dbm$$

$$\approx \boxed{-8.24\ dbm}$$

b) How many dBm is 0.025 watts?

$$0.025\ W = 25 \cdot \underbrace{10^{-3}\ W}$$

$$25\ mW$$

$$P_{dbm} = 10\log_{10}(25) = 13.97940\ dbm \approx 13.98\ dbm$$

c) How many watts is $-30\ dBm$ (provide your answer in W)?

$$P_{mW} = 10^{\frac{-30}{10}} = 10^{-3}\ mW$$

$$= 10^{-3} \cdot 10^{-3}\ W \qquad \boxed{= 10^{-6}\ W}$$

d) How many volts-~~rms~~ is across a 50Ω resistor dissipating $-10\ dBm$ of power?

$$P_{mW} = 10^{\frac{-10}{10}} = 10^{-1}\ mW = 10^{-1} \cdot 10^{-3}\ W$$

$$= 10^{-4}\ W = 0.0001 W$$

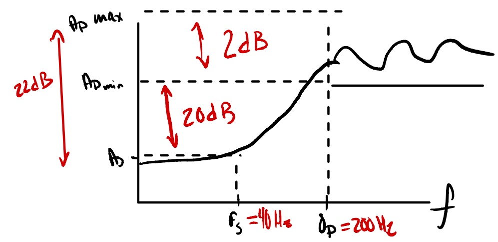
$$P = \frac{V^2}{R} \rightarrow V = \sqrt{P \cdot R}$$

$$= \sqrt{(0.0002 \cdot 50} = 0.07071 W \qquad \approx \boxed{70.7 mV}$$

lecture 6 Panopto

Lecture 5b slide #15:

See CMRR examples



Lecture 10 and identifying filter order:

See Notes on Lecture 10: Filter Specifications

High-Pass Filter Specifications:

a) The passband ripple α_p should be less than or equal to 2 dB (i.e., $A_{p,max} - A_{p,min} \leq 2$ dB).
b) The desired low-frequency passband corner frequency is $f_{p,L} = 200$ Hz.
c) The desired low-frequency stopband corner frequency is $f_{s,L} = 40$ Hz.
d) The stopband must have an attenuation of at least 22 dB below $A_{p,max}$.

$$\left|\frac{A_{p,min} - A_s}{\log_{10}(f_s) - \log_{10}(f_p)}\right| \leq 20n$$

$$\frac{\left|\dfrac{20}{\log_{10}(40) - \log_{10}(200)}\right|}{20} = 1.43 \leq n$$

$$\boxed{\therefore n = 2}$$